

ALLIANCE UNIVERSITY

Private University established in Karnataka State by Act No.34 of year 2010
Recognized by the University Grants Commission (UGC), New Delhi

NAAC GRADE A+ ACCREDITED UNIVERSITY

Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 2

POWER BI PROJECT REPORT

Title: SUPERSTORE SALES PERFORMANCE ANALYSIS

Submitted By

Name: Prem Panchal

Reg. No.: 2411021240011

Date: 31/08/2026

Submitted To

Faculty Name: Mr. Aman Kumar Sharma

Department of Computer Application

Alliance University

Chandapura - Anekal Main Road, Anekal

Bengaluru – 562106

1. Introduction

This project uses the Sample Superstore business dataset to develop an interactive sales-performance report in Microsoft Power BI. The objective is to help management quickly understand sales performance across products, categories, and regions and to identify important business patterns.

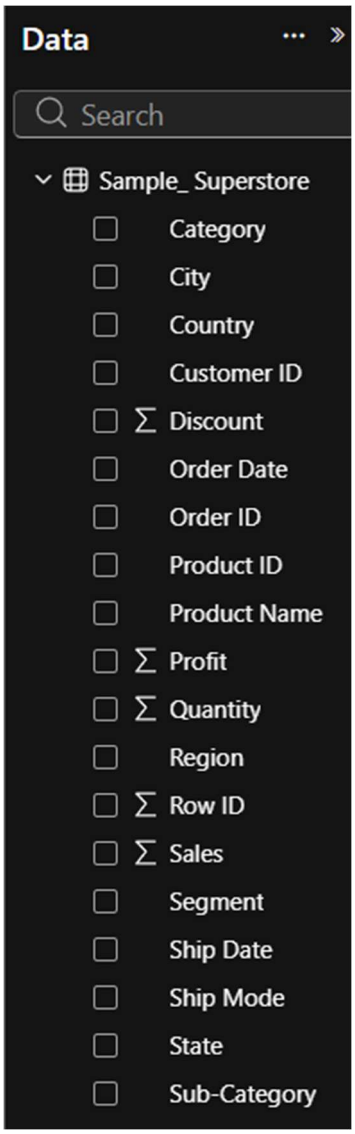
2. Problem Statement

Management wants to quickly understand sales performance across products, categories, and regions. Create appropriate Power BI visualizations to identify key business patterns.

3. Dataset Used

The project uses the Sample Superstore dataset specified in the assignment. The imported table contains business fields covering orders, products, customers, locations, sales, profit, quantity, discount, and dates.

- Main categorical fields used: Category, Sub-Category, Region, State, and Product Name.
- Main numerical fields used: Sales, Profit, Quantity, and Discount.
- Date field used for trend analysis: Order Date.



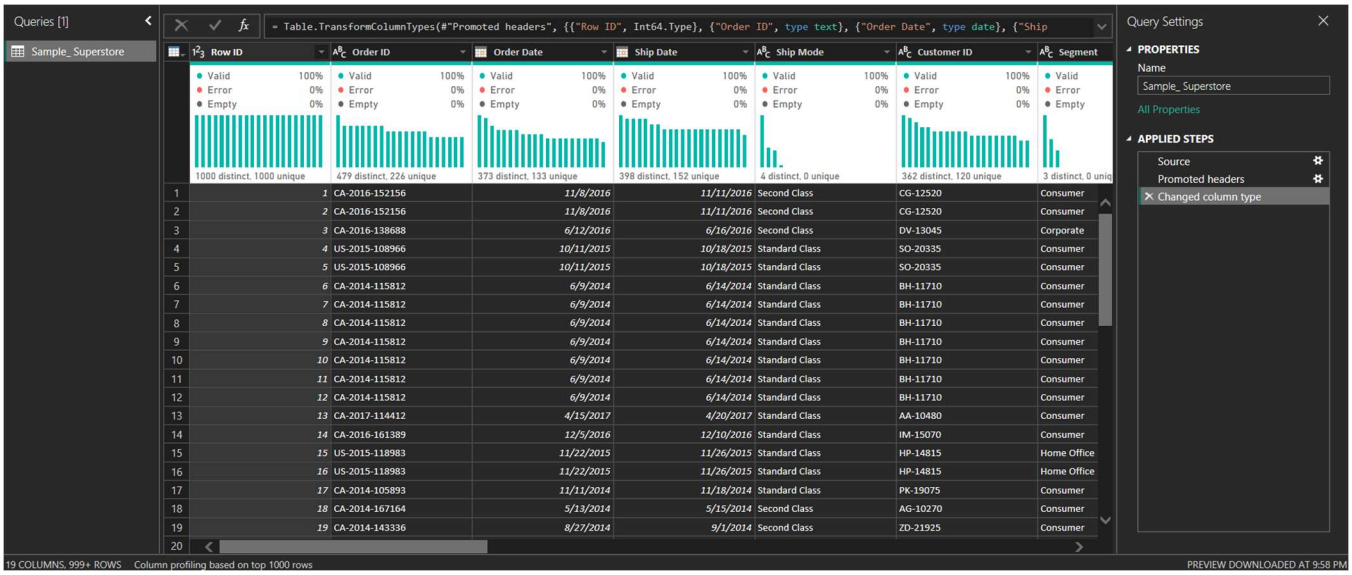
4. Tools and Technologies

- Microsoft Power BI Desktop
- Power Query for data preparation
- Sample Superstore CSV dataset
- Power BI visualizations, slicers, and dashboard pages

5. Data Import and Preparation

The Sample Superstore CSV file was imported into Power BI Desktop using the Get Data option. The data was checked in Power Query and prepared for reporting. The Order Date field was used as the date field for time-based analysis.

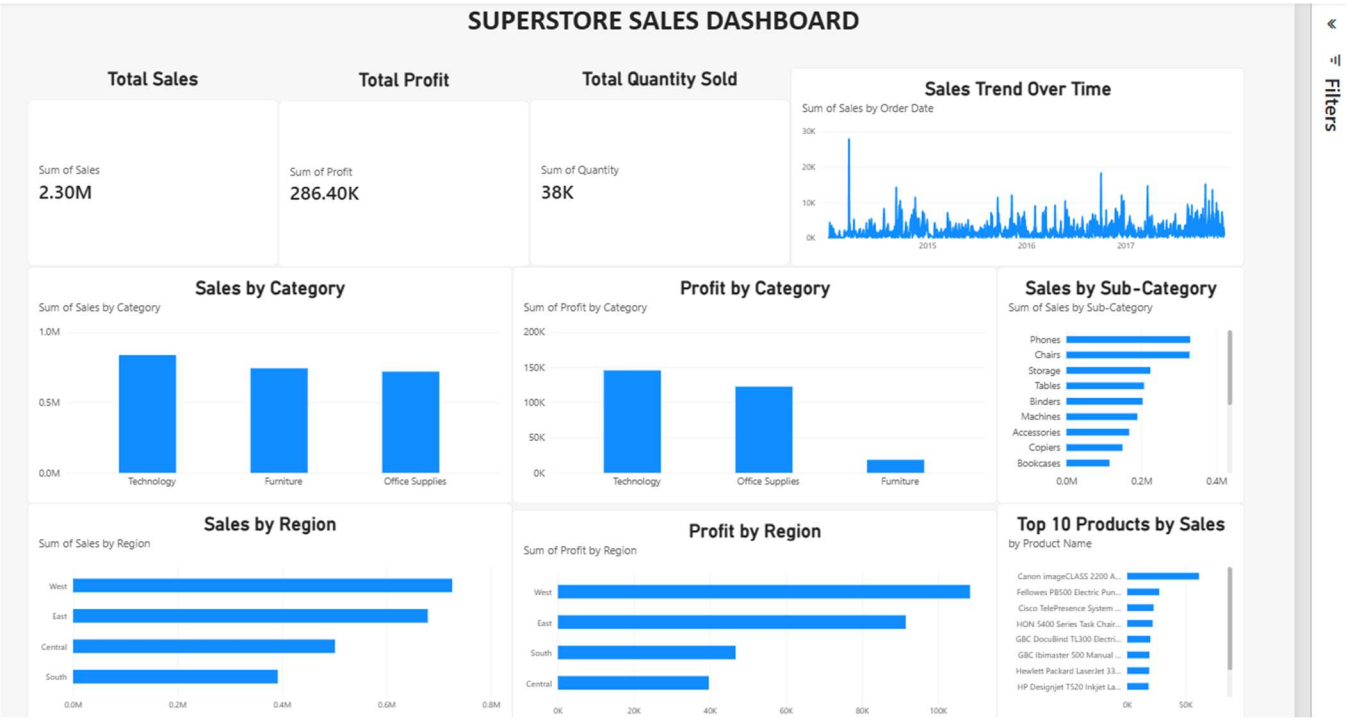
- Imported the Sample Superstore CSV into Power BI.
- Verified the available fields and data types.
- Used Power Query for the data preparation stage.
- Used Sales, Profit, and Quantity as the main numerical measures.
- Used Order Date for sales trend analysis.



6. Dashboard Development

The main report page was designed as a Sales Performance Dashboard. It provides a centralized view of the most important sales and profitability measures and allows management to compare categories, regions, sub-categories, and products.

- Created KPI cards for Total Sales, Total Profit, and Total Quantity Sold.
- Created a Sales by Category column chart.
- Created a Profit by Category column chart.
- Created a Sales by Sub-Category horizontal bar chart.
- Created a Sales by Region horizontal bar chart.
- Created a Profit by Region horizontal bar chart.
- Created a Top 10 Products by Sales bar chart.
- Created a Sales Trend Over Time line chart.



7. Filters and Analysis Page

A separate Filters & Analysis page was created to keep the main dashboard clean while providing interactive filtering and additional analysis.

- Category slicer for Furniture, Office Supplies, and Technology.
- Region slicer for Central, East, South, and West.
- Order Date filter for selecting a date range.
- Sales by State map for geographical analysis.
- Sales vs Profit by Sub-Category scatter chart for profitability analysis.

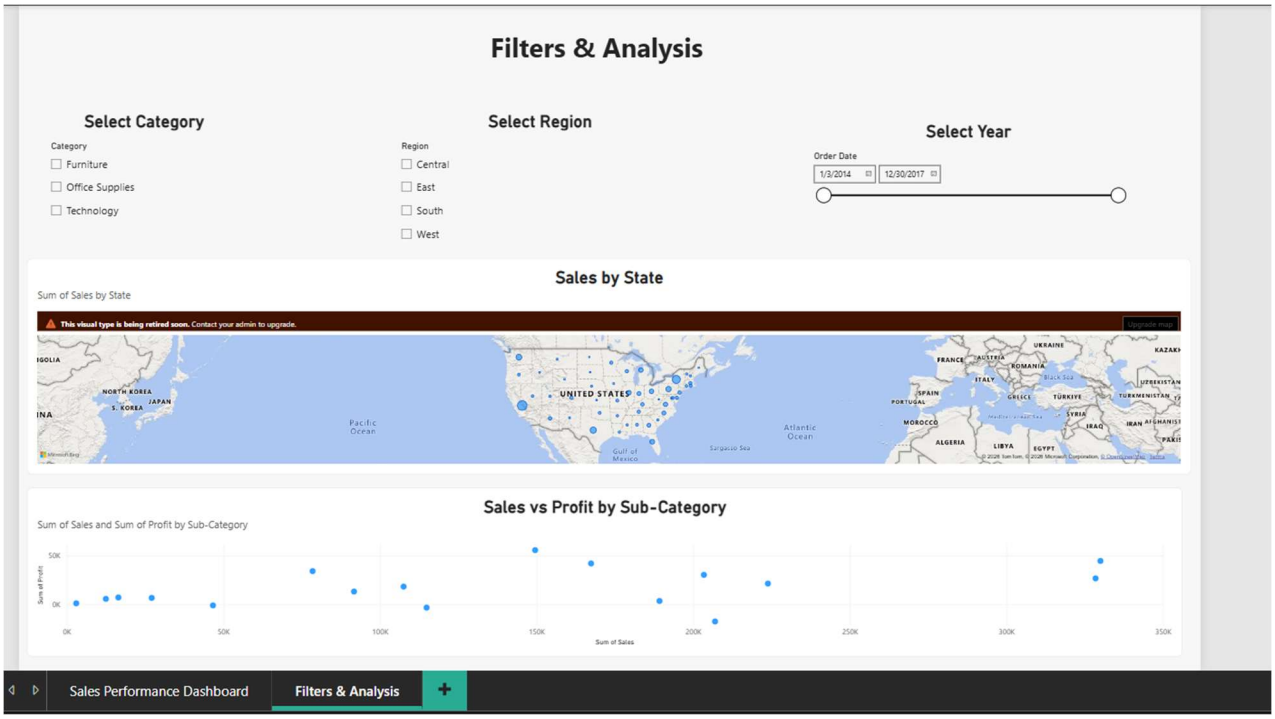


Figure 3: Filters & Analysis page

8. Key Business Findings

- Overall performance: the dashboard shows approximately 2.30M in total sales, 286.40K in total profit, and 38K in total quantity sold.
- Category performance: Technology has the highest sales among the three categories and also the highest profit contribution in the report.
- Category profitability: Furniture has substantially lower profit than Technology and Office Supplies, despite contributing significant sales.
- Regional performance: West has the highest sales and profit among the four regions, followed by East.
- Sub-category performance: Phones and Chairs are among the leading sub-categories by sales in the displayed report.
- Product performance: the Top 10 Products by Sales visual identifies major individual revenue contributors.
- Time analysis: the sales trend visual shows fluctuations over the available Order Date period and helps identify periods of higher and lower sales activity.

9. Business Use of the Dashboard

- Management can compare sales and profit across categories and regions.
- High-performing products can be identified for closer business attention.
- Regional differences can support sales planning and resource allocation.
- Interactive filters allow analysis by selected category, region, or date range.
- The dashboard provides a centralized visual summary for faster data-driven decision-making.

10. Conclusion

The Superstore Sales Performance Analysis project successfully transforms the Sample Superstore dataset into an interactive Power BI report. The dashboard combines KPI cards, categorical comparisons, regional analysis, product rankings, time-series analysis, filters, and profitability analysis. These visualizations provide management with a clear and centralized view of sales performance and help identify important business patterns for decision-making.

11. Learning Outcomes

- Learned how to import a CSV business dataset into Power BI.
- Learned how to inspect and prepare business data using Power Query.
- Learned how to create KPI cards and business visualizations.
- Learned how to analyze sales by category, sub-category, product, and region.
- Learned how to create interactive slicers and filters.
- Learned how to compare sales and profit to identify business patterns.
- Learned how to design a structured Power BI dashboard for management reporting.

12. Output

The final Power BI project consists of two report pages: Sales Performance Dashboard and Filters & Analysis. Together, these pages provide an interactive view of overall sales, profitability, products, categories, regions, states, and sales trends.